Assignment 4

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Batch: B6

GitHub link: https://github.com/pankhuri007-007/Pankhuri_590032857_B6

Course: B.Tech. CSE (Sem I)

Subject: Programming in C

1. **Data Type Size and Range**:

```
compilation terminated.
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> gcc assign4.c
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> ./a.exe
Size of char = 1
Size of int = 4
Size of float = 4
Size of double = 8
Size of long int = 4
Size of short int = 2

Minimum int = -2147483648
Maximum int = 2147483647
Minimum char = -128
Maximum char = 127
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code>
```

2. Type Conversion in Expressions:

```
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> gcc assign4.2.c
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> ./a.exe
Enter an integer: 15
Enter a floating-point number: 7.7

Implicit Type Conversion:
Addition = 22.700000
Subtraction = 7.300000
Multiplication = 115.499997
Division = 1.948052

Explicit Type Conversion:
Addition = 22.700000
Subtraction = 7.300000
Multiplication = 115.499997
```

```
Enter an integer: 15
Enter a floating-point number: 7.7

Implicit Type Conversion:
Addition = 22.700000
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Explicit Type Conversion:
Addition = 22.700000
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Division = 1.948052
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code>
```

3. Variable Scope:

```
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> gcc assign4.3.c
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> ./a.exe
Inside main before block:
Global variable a = 10
Local variable b = 20
Static variable c = 30

Inside block:
Global variable a = 15
Block local variable b = 55
Static variable c = 35

Inside main after block:
Global variable a = 15
```

```
Inside main before block:
Global variable a = 10
Local variable b = 20
Static variable c = 30

Inside block:
Global variable a = 15
Block local variable b = 55
Static variable c = 35

Inside main after block:
Global variable a = 15
Local variable b = 20
Static variable c = 35
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code>
```

4. Keywords Identification:

```
Problems    Output    Debug Console    Terminal    Ports

dence" for more details.
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code>
 *  History restored

PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> gcc assign4.4.c
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> ./a.exe
Enter marks obtained out of 500: 380
Very Good
Percentage = 76.00
Grade = B
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code>
```

Keywords	Purpose
1.int	Used to declare integer variables
2.float	Used to declare floating-point variables.
3.char	Used to declare character variables.

4.const	**Used to declare a constant value**
5.if	**Used to check a condition**
6.else	**Used when the if condition is false.**
7.return	**Used to return a value from the function**

5. Arithmetic and Mathematical Operators Using Header File:

```
Problems    Output    Debug Console    Terminal    Ports

PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> gcc assign4.5.c
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> ./a.exe
Enter two integer numbers: 67
56
Addition = 123
Subtraction = 11
Multiplication = 3752
Division = 1.20
Modulus = 11
Power = 1820492739620332800000000000000000000000000000000000000000000000000000000000000000000000000000000000000.00
Square root of a = 8.19
Absolute value of a = 67
Floor value of a = 67.00
Ceiling value of a = 67.00
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code>
```

6. Relational and Logical Operators:

- **Passed all subjects:**

```
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> gcc assign4.6.c
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> ./a.exe
Enter marks of three subjects: 40
50
78
Student passed all subjects
Student did not score distinction
Average marks = 56.00
```

- **Failed in one or more subjects:**

```
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> gcc assign4.6.c
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> ./a.exe
Enter marks of three subjects: 36
36
50
Student failed in one or more subjects
Average marks = 40.67
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code>
```

- **Scored distinction if average is 75 or above**

```
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> gcc assign4.6.c
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> ./a.exe
Enter marks of three subjects: 88
78
60
Student passed all subjects
Student scored distinction
Average marks = 75.33
```

7. Increment and Decrement Operators:

```
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> gcc assign4.7.c
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> ./a.exe
Initial value of a = 10
Using ++a = 11
Value of a after ++a = 11

Initial value of a = 10
Using a++ = 10
Value of a after a++ = 11

Initial value of a = 10
Using --a = 9
Value of a after --a = 9

Initial value of a = 10
```

```
Problems    Output    Debug Console    Terminal    Ports

Using ++a = 11
Value of a after ++a = 11

Initial value of a = 10
Using a++ = 10
Value of a after a++ = 11

Initial value of a = 10
Using --a = 9
Value of a after --a = 9

Initial value of a = 10
Using a-- = 10
Value of a after a-- = 9
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code>
```

8. Complex Logical Condition:

- **age is between 21 and 60:**
- **income is greater than 35000**
- **credit score is greater than or equal to 750**

```
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> gcc assign4.8.c
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> ./a.exe
Enter age: 18
Enter income: 100000
Enter credit score: 600
Person is not eligible for loan
```

```
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> gcc assign4.8.c
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> ./a.exe
Enter age: 21
Enter income:

3500
Enter credit score: 750
Person is not eligible for loan
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> gcc assign4.8.c
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> ./a.exe
Enter age: 27
Enter income: 70000
Enter credit score: 850
Person is eligible for loan
```

9. Operator Precedence:

```
Operator precedence is verified
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code>  gcc assign4.9.c
PS C:\Users\Pankhuri Khanna\Desktop\UPES\Programming UPES\Code> ./a.exe
Enter values of a, b, c, d, e, f, g:  10 2 3 20 5 17 4
Result1 = 13
Result2 = 13
Operator precedence is verified
```